

Latent Diffusion for High-Resolution Image Synthesis

Abstract

We present a latent diffusion model generating 1024px images with FID score 4.2.

Introduction

Diffusion models produce high-quality images but require massive compute.

Method

Operating in latent space reduces compute by 85% while preserving quality.

Experiment

FID 4.2 on COCO, IS 245.6, user study preference 78% over baseline.

Conclusion

Latent diffusion bridges quality and efficiency for image generation.